Recognition of Linear and Star Variants of Leaf Powers is in P

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Abstract. A k-leaf power of a tree T is a graph G whose vertices are the leaves of T and whose edges connect pairs of leaves whose distance in T is at most k. A graph is a leaf power if it is a k-leaf power for some k. Over 20 years ago, Nishimura et al. [J. Algorithms, 2002] asked if recognition of leaf powers was in P. Recently, Lafond [SODA 2022] showed an XP algorithm when parameterized by k, while leaving the main question open. In this paper, we explore this question from the perspective of two alternative models of leaf powers, showing that both a linear and a star variant of leaf powers can be recognized in polynomial-time.

Keywords: Leaf power · Co-threshold tolerance graphs · Interval graphs

1 Introduction

Leaf powers were introduced by Nishimura et al. in [22], and have enjoyed a steady stream of research. Leaf powers are related to the problem of reconstructing *phylogenetic trees*. For an integer k, a graph G is a k-leaf power if there exists a tree T – called a *leaf root* – with a one-to-one correspondence between $V(G)$ and the leaves of T, such that two vertices u and v are neighbors in G iff the distance between the two corresponding leaves in T is at most k. G is a leaf powers if it is a k-leaf power for some k. The most important open problem in the field is whether leaf powers can be recognized in polynomial time.

Most of the results on leaf powers have followed two main lines, focusing either on the distance values k or on the relation of leaf powers to other graph classes, see e.g. the survey by Calamoneri et al. [7]. For the first approach, steady research for increasing values of k has shown that k-leaf powers for any $k \leqslant 6$ is recognizable in polytime [4,5,8,11,12,22]. Moreover, the recognition of k-leaf powers is known to be FPT parameterized by k and the degeneracy of the graph [13]. Recently Lafond [19] gave a polynomial time algorithm to

Omitted proofs and a conclusion can be found in the full version of this paper available on https://arxiv.org/abs/2105.12407.

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recognize k-leaf powers for any constant value of k. For the second approach, we can mention that interval graphs [5] and rooted directed path graphs [3] are leaf powers, and also that leaf powers have mim-width one [17] and are strongly chordal. Moreover, an infinite family of strongly chordal graphs that are not leaf powers has been identified [18]; see also Nevris and Rosenke [21].

To decide if leaf powers are recognizable in polynomial time, it may be better not to focus on the distance values k. Firstly, the specialized algorithms for k-leaf powers for $k \leqslant 6$ do not seem to generalize. Secondly, the recent XP algorithm of Lafond [19] uses techniques that will not allow removing k from the exponent. In this paper we therefore take a different approach, and consider alternative models for leaf powers that do not rely on a distance bound. In order to make progress towards settling the main question, we consider fundamental restrictions on the shape of the trees, in two distinct directions: subdivided caterpillars (linear) and subdivided stars, in both cases showing polynomial-time recognizability. We use two models: weighted leaf roots for the linear case and NeS models for the stars.

The first model uses rational edge weights between 0 and 1 in the tree T which allows to fix a bound of 1 for the tree distance. It is not hard to see that this coincides with the standard definition of leaf powers using an unweighted tree T and a bound k on distance. Given a solution of the latter type we simply set all edge weights to $1/k$, while in the other direction we let k be the least common denominator of all edge weights and then subdivide each edge a number of times equal to its weight times k.

The second model arises by combining the result of Brandstädt et al. that leaf powers are exactly the fixed tolerance NeST graphs [3, Theorem 4] with the result of Bibelnieks et al. [1, Theorem 3.3] that these latter graphs are exactly those that admit what they call a "neighborhood subtree intersection representation", that we choose to call a NeS model. NeS models are a generalization of interval models: by considering intervals of the line as having a center that stretches uniformly in both directions, we can generalize the line to a tree embedded in the plane, and the intervals to embedded subtrees with a center, stretching uniformly in all directions from the center, along tree edges. Thus a NeS model of a graph G consists of an embedded tree and one such subtree for each vertex, such that two vertices are adjacent in G iff their subtrees have non-empty intersection. Precise definitions are given later. The leaf powers are exactly the graphs having a NeS model. Some results are much easier to prove using NeS models, to illustrate this, we show that leaf powers are closed under several operations such as the addition of a universal vertex (see Lemma 6).

We show that fundamental constraints on these models allow polynomial-time recognition. Using the first model, we restrict to edge-weighted caterpillars, i.e. trees with a path containing all nodes of degree 2 or more. We call linear leaf power a graph with such model. Brandstädt et al. [2] considered leaf roots restricted to caterpillars (see also [6]) in the unweighted setting, showing that unit interval graphs are exactly the k-leaf powers for some k with a leaf root being an unweighted caterpillar. In the unweighted setting, linear leaf powers are graphs with a leaf root that is a *subdivision* of a caterpillar. We show that linear leaf powers are exactly the *co-threshold tolerance graphs* [20], and combined with the algorithm of Golovach et al. [14] this implies that we can recognize linear leaf powers in $O(n^2)$ time. Our proof goes via the equivalent concept of blue-red interval graphs introduced by [15].

The recognition of linear leaf powers in polynomial time could have practical applications for deciding whether the most-likely evolutionary tree associated with a set of organisms has a linear topology. Answering this question might find particular relevance inside the field of tumor phylogenetics where, under certain model assumptions, linear topologies are considered more likely [9,24].

For NeS models, we restrict to graphs having a NeS model where the embedding tree is a star, and show that they can be recognized in polynomial time. Note that allowing the embedding tree to be a subdivided star will result in the same class of graphs. Our algorithm uses the fact that the input graph must be a chordal graph, and for each maximal clique X we check if G admits a star NeS model where the set of vertices having a subtree containing the central vertex of the star is X. To check this we use a combinatorial characterization, that we call a "good partition", of a star NeS model.

2 Preliminaries

For positive integer k, denote by $[k]$ the set $\{1, 2, \dots, k\}$. A partition of a set S is a collection of non-empty disjoint subsets $B_1, \dots, B_t$ of S – called *blocks* – such that $S = B_1 \cup \dots \cup B_t$. Given two partitions $\mathcal{A}, \mathcal{B}$ of S, we say $\mathcal{A} \sqsubseteq \mathcal{B}$ if every block of $\mathcal{A}$ is included in a block of $\mathcal{B}$, i.e. $\sqsubseteq$ is the *refinement* relation.

Graph. Our graph terminology is standard and we refer to [10]. The vertex set of a graph G is denoted by $V(G)$ and its edge set by $E(G)$. An edge between two vertices x and y is denoted by xy or yx. The set of vertices that is adjacent to x is denoted by $N(x)$. A vertex x is simplicial if $N(x)$ is a clique. Two vertices x, y are true twins if $xy \in E(G)$ and $N(x)\backslash\{y\} = N(y)\backslash\{x\}$. Given $X \subseteq V(G)$, we denote by $G[X]$ the graph induced by X. Given a vertex $v \in V(G)$, we denote the subgraph $G[V(G)\backslash\{v\}]$ by $G - v$. We denote by $\mathsf{CC}(G)$ the partition of G into its connected components. Given a tree T and an edge-weight function $\mathsf{w} : E(T) \to \mathbb{Q}$, the distance between two vertices x and y is denoted by $d_T(x, y)$ is $\sum_{e \in E(P)} \mathsf{w}(e)$ with P is the unique path between x and y. A caterpillar is a tree in which there exists a path that contains every vertex of degree two or more

Leaf Power. In the Introduction we have already given the standard definition of leaf powers and leaf roots, and also we argued the equivalence with the following. Given a graph G, a leaf root of G is a pair (T, w) of a tree T and a rational-valued weight function $\mathsf{w} : E(T) \to [0, 1]$ such that the vertices of G are the leaves of T and for every $u, v \in V(G)$, u and v are adjacent iff $d_T(u, v) \leqslant 1$. Moreover, if T is a caterpillar we call (T, w) a *linear leaf root*. A graph is a *leaf power* if it admits a leaf root and it is a linear leaf power if it admits a linear leaf root. Since we manipulate both the graphs and the trees representing them, the vertices of trees will be called *nodes* to avoid confusion.

Interval Graphs. A graph G is an *interval graph* if there exists a set of intervals in $\mathbb{Q}$, $\mathcal{I} = (I_v)_{v \in V(G)}$, such that for every pair of vertices $u, v \in V(G)$, the intervals I_v and I_u intersect iff $uv \in E(G)$. We call $(I_v)_{v \in V(G)}$ an *interval model* of G. For an interval $I = [\ell, r]$, we define the midpoint of I as $(\ell + r)/2$ and its length as $r - \ell$.

Clique Tree. A Chordal graph is a graph in which every induced cycle have exactly three vertices. For a chordal graph G, a clique tree CT of G is a tree whose vertices are the maximal cliques of $V(G)$ and for every vertex $v \in V(G)$, the set of maximal cliques of G containing v induces a subtree of CT. Figure 2 gives an example of clique tree. Every chordal graph admits $O(n)$ maximal cliques and given a graph G, in time $O(n+m)$ we can construct a clique tree of G or confirm that G is not chordal [16,23]. When a clique tree is a path, we call it a clique path. We denote by $(K_1, \ldots, K_k)$ the clique path whose vertices are $K_1, \ldots, K_k$ and where K_i is adjacent to K_{i+1} for every $i \in [k-1]$.

3 Linear Leaf Powers

In this section we show that linear leaf powers are exactly the *co-threshold tolerance graphs* (co-TT graphs). Combined with the algorithm in [14], this implies that we can recognize linear leaf powers in $O(n^2)$ time.

Co-TT graphs were defined by Monma, Trotter and Reed in [20]; we will not define them here as we do not use this characterization. Rather, we work with the equivalent class of *blue-red interval graphs* [15, Proposition 3.3].

Definition 1 (Blue-red interval graph). *A graph G is a* blue-red interval graph *if there exists a bipartition (B, R) of $V(G)$ and an interval model $\mathcal{I} = (I_v)_{v \in V(G)}$ (with $(B, R, \mathcal{I})$ called a blue-red interval model) such that $E(G) = \{b_1 b_2 : b_1, b_2 \in B$ and $I_{b_1} \cap I_{b_2} \neq \varnothing\} \cup \{rb : r \in R, b \in B$ and $I_r \subseteq I_b\}$.*

The red vertices induce an independent set, $(I_b)_{b \in B}$ is an interval model of $G[B]$, and we have a blue-red edge for each red interval contained in a blue interval. The following fact can be easily deduced from Fig. 1.

Fact 2 *Consider two intervals I_1, I_2 with lengths ℓ_1, ℓ_2 and midpoints m_1, m_2 respectively. We have $I_1 \cap I_2 \neq \varnothing$ iff $|m_1 - m_2| \leqslant \frac{\ell_1 + \ell_2}{2}$. Moreover, we have $I_2 \subseteq I_1$ iff $|m_1 - m_2| \leqslant \frac{\ell_1 - \ell_2}{2}$.*

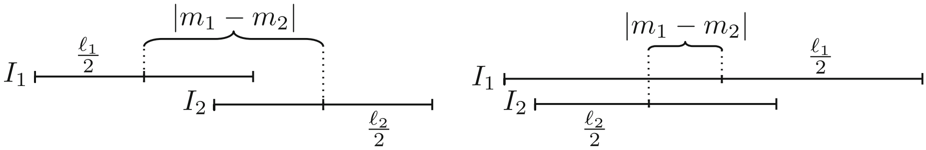

Fig. 1. Example of two intervals overlapping and one interval containing another one.

To prove that linear leaf powers are exactly blue-red interval graphs, we use a similar construction as the one used in [2, Theorem 6] to prove that every interval graph is a leaf power, but in our setting, we have to deal with red vertices and this complicates things quite a bit.

Theorem 3. *G is a blue-red interval graph iff G is a linear leaf power.*

Proof (Sketch of proof). ($\Rightarrow$) Let $(B, R, (I_v)_{v \in V(G)})$ be a blue-red interval model of a graph G. For each $v \in V(G)$, we denote by m_v and ℓ_v the midpoint and the length of the interval I_v. We assume w.l.o.g. that G is connected and the lengths of the intervals I_v's are not 0 and at most 1. We prove that we can construct a linear leaf root (T, w) of G. For doing so, we construct (T, w) as follows. The inner path of T represents the midpoints of the intervals: each node of this path is associated with a midpoint and the weights on the edges of this path are the distances between consecutive midpoints (since G is connected by assumption and the length are at most 1, these distances are at most 1). Each vertex v of G is adjacent through an edge e to the node representing the midpoint m_v, the weight of e is $\frac{1-\ell_v}{2}$ if $v \in B$ and $\frac{1+\ell_v}{2}$ if $v \in R$. With these weights, the distance between two red vertices in T is strictly greater than 1. Moreover, for every $v \in B$ and $u \in V(G)$, we have (♠) $d_T(u, v) = 1 + |m_u - m_v| - \left(\frac{\ell_u+\ell_v}{2}\right)$ if $u \in B$ and $1 + |m_v - m_u| - \left(\frac{\ell_v-\ell_u}{2}\right)$ if $u \in R$. From Fact 2, one easily proves that (T, w) is a linear leaf root of G.

($\Leftarrow$) Let (T, w) be a linear leaf root of a graph G with $(u_1, \ldots, u_t)$ the path induced by the internal vertices of T. The previous construction can easily be reversed to construct a blue-red interval model $(B, R, (I_v)_{v \in V(G)})$ of G. For each vertex $v \in V(G)$ whose neighbor in T is u_i, we associate v with an interval I_v and a color as follows. We color v blue if $\mathsf{w}(u_i v) \leqslant 1/2$, otherwise we color it red. The midpoint m_v of I_v is the distance between u_1 and u_i in T. We define the length ℓ_v of I_v to be $1 - 2\mathsf{w}(u_i v)$ if v is blue and $2\mathsf{w}(u_i v) - 1$ if v is red. By construction, $\mathsf{w}(u_i v) = \frac{1-\ell_v}{2}$ if $v \in R$ and $\frac{1+\ell_v}{2}$ if $v \in B$. Observe that two red vertices cannot be adjacent since their distance in T is strictly greater than 1. We can prove that (♠) holds also for this direction and with Fact 2 one concludes that $(B, R, (I_v)_{v \in V(G)})$ is a blue-red interval model of G. $\square$

4 Star NeS Model

In this section, we first present an alternative definition of leaf powers through the notion of NeS models. We then show that we can recognize in polynomial time graphs with a *star NeS model*: a NeS model whose embedding tree is a star (considering subdivided stars instead of stars does not make a difference).

For each tree T, we consider a corresponding tree $\mathcal{T}$ embedded in the Euclidean plane so that each edge of T corresponds to a line segment of $\mathcal{T}$, these lines segments can intersect one another only at their endpoints, and the vertices of T correspond (one-to-one) to the endpoints of the lines. Each line segment of $\mathcal{T}$ has a positive Euclidean length. These embedded trees allow us to consider $\mathcal{T}$ as the infinite set of points on the line segments of $\mathcal{T}$. The notion of tree embedding used here is consistent with that found in [25]. The line segments of $\mathcal{T}$ and their endpoints are called respectively the edges and the nodes of $\mathcal{T}$. The distance between two points x, y of $\mathcal{T}$ denoted by $d_{\mathcal{T}}(x, y)$ is the length of the unique path in $\mathcal{T}$ between x and y (the distance between two vertices of T and their corresponding endpoints in $\mathcal{T}$ are the same).

Definition 4 (Neighborhood subtree, NeS-model). *Let $\mathcal{T}$ be an embedding tree. For some point $c \in \mathcal{T}$ and non-negative rational w, we define the neighborhood subtree with center c and radius w as the set of points $\{p \in \mathcal{T} : d_{\mathcal{T}}(p, c) \leqslant w\}$. A NeS model $(\mathcal{T}, (T_v)_{v \in V(G)})$ of a graph G is a pair of an embedding tree $\mathcal{T}$ and a collection of neigbhorhood subtrees of $\mathcal{T}$ associated with each vertex of G such that for every $u, v \in V(G)$, we have $uv \in E(G)$ iff $T_u \cap T_v \neq \varnothing$.*

Theorem 5. *A graph is a leaf power iff it admits a NeS model.*

Proof. Brandstädt et al. showed that leaf powers correspond to the graph class fixed tolerance NeST graph [3, Theorem 4]. Bidelnieks and Dearing showed that G is a fixed tolerance NeST graph iff G has a NeS model [1, Theorem 3.3]. □

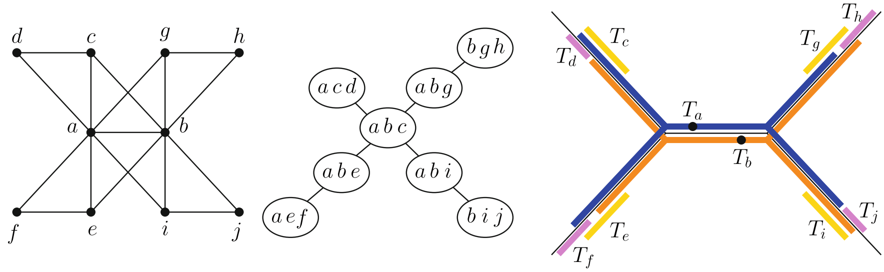

Fig. 2. A graph with a clique tree and NeS model. The dots are centers of T_a and T_b.

See Fig. 2 for a NeS model. From now, we consider that the intervals of an interval models are segments of a line in the plane rather than intervals in $\mathbb{Q}$. Observe that every interval graph has a NeS model where the embedding tree is a single edge. Moreover, if a graph G admits a NeS model $(\mathcal{T}, (T_v)_{v \in V(G)})$, then for every embedding path $\mathcal{L}$ of $\mathcal{T}$, $(\mathcal{L}, (T_v \cap \mathcal{L})_{v \in X})$ is an interval model of $G[X]$ with X the set of vertices v such that T_v intersects $\mathcal{L}$. As illustrated by the proofs of the following two lemmata, some results are easier to prove with NeS models than with other characterizations.

Lemma 6. *For a graph G and $u \in V(G)$ such that either (1) u is universal, or (2) u has degree 1, or (3) $N(u)$ is a minimal separator in $G - u$, or (4) $N(u)$ is a maximal clique in $G - u$. Then G is a leaf power iff $G - u$ is a leaf power.*

Lemma 7. *Let G be a graph and $u \in V(G)$ a cut vertex. Then G is a leaf power iff for every component C of $G - u$, $G[V(C) \cup \{u\}]$ is a leaf power.*

We now give the algorithm for recognizing graphs having a star NeS model. Our result is based on the purely combinatorial definition of *good partition*, and we show that a graph admits a star NeS model iff it admits a good partition. Given a good partition, we compute a star NeS model in polynomial time. Finally, we prove that our Algorithm 1 in polynomial time constructs a good partition of the input graph or confirms that it does not admit one.

Consider a star NeS model $(\mathcal{T}, (T_v)_{v\in V(G)})$ of a graph G. Observe that $\mathcal{T}$ is the union of line segments $L_1, \ldots, L_\beta$ with a common endpoint c that is the center of $\mathcal{T}$. Let X be the set of vertices whose neighborhood subtrees contain c. For each $i \in [\beta]$, we let B_i be the set of all vertices in $V(G)\backslash X$ whose neighborhood subtrees are subsets of L_i. The family $\mathcal{B} = \{B_i : i \in [\beta]\}$ must then constitute a partition of $V(G)\backslash X$. We will show in Theorem 12 that the pair $(X, \mathcal{B})$ has the properties of a good partition.

Fact 8 *Let G, $(\mathcal{T}, (T_v)_{v\in V(G)})$ and $\mathcal{B}$ be as defined above. We then have:*

- *There is no edge between B_i and B_j for $i \neq j$ and thus $\mathsf{CC}(G - X) \sqsubseteq \mathcal{B}$.*
- *For every $i \in [\beta]$ the NeS model $(L_i, (T_v \cap L_i)_{v\in B_i\cup X})$ is an interval model of $G[X \cup B_i]$.*
- *For each $x \in X$ the neighborhood subtree T_x is the union of the β intervals $L_1 \cap T_x, \ldots, L_\beta \cap T_x$ and there exist positive rationals ℓ_x and h_x with $\ell_x \leqslant h_x$ such that one interval among these intervals has length h_x and the other $\beta-1$ intervals have length ℓ_x. If $\ell_x = h_x$, then the center of T_x is c.*

Claim. If G has a star NeS model, it has a one, $(\mathcal{T}, (T_v)_{v\in V(G)})$, where vertices whose neighborhood subtrees contain the center of $\mathcal{T}$ is a maximal clique.

Claim 4 follows since we can always stretch some intervals to make X a maximal clique. So far we have described a good partition as it arises from a star NeS model. Now we introduce the properties of a good partition that will allow to abstract away from geometrical aspects while still being equivalent, i.e. so that a graph has a good partition $(X, \mathcal{B})$ iff it has a star NeS model. The first property is $\mathsf{CC}(G - X) \sqsubseteq \mathcal{B}$ and the second is that for every $B \in \mathcal{B}$ the graph $G[X \cup B]$ is an interval graph having a model where the intervals of X contain the last point used in the interval representation.

Definition 9 (X-interval graph). *Let X be a maximal clique of G. We say that G is an X-interval graph if G admits a clique path ending with X.*

The third property is the existence of an elimination order for the vertices of X based on the lengths ℓ_x in the last item of Fact 8, namely the permutation $(x_1, \ldots, x_t)$ of X such that $\ell_{x_1} \leqslant \ell_{x_2} \leqslant \ldots \leqslant \ell_{x_t}$. This permutation has the property that for any $i \in [t]$, among the vertices $x_i, x_{i+1}, ..., x_t$ the vertex x_i must have the minimal neighborhood in at least $\beta - 1$ of the blocks of $\mathcal{B}$; we say that x_i is *removable* from $\{x_i, \ldots, x_t\}$ for $\mathcal{B}$.

Definition 10 (Removable vertex). *Let $X \subseteq V(G)$, $Y \subseteq X$ and let $\mathcal{B}$ be a partition of $V(G)\backslash X$. Given a block B of $\mathcal{B}$ and $x \in Y$, we say $N(x)$ is minimal in B for Y if $N(x) \cap B \subseteq N(y)$ for every $y \in Y$. We say that a vertex $x \in Y$ is removable from Y for $\mathcal{B}$ if $N(x)$ is minimal in at least $|\mathcal{B}| - 1$ blocks of $\mathcal{B}$ for Y.*

Definition 11 (Good partition). *A good partition of a graph G is a pair $(X, \mathcal{B})$ where X is a maximal clique of G and $\mathcal{B}$ a partition of $V(G) \setminus X$ satisfying:*

1. $\mathsf{CC}(G - X) \sqsubseteq \mathcal{B}$, *i.e. every $C \in \mathsf{CC}(G - X)$ is contained in a block of $\mathcal{B}$.*
2. *For each block $B \in \mathcal{B}$, $G[X \cup B]$ is an X-interval graph.*
3. *There exists an elimination order $(x_1, \ldots, x_t)$ on X such that for every $i \in [t]$, x_i is removable from $\{x_i, \ldots, x_t\}$ for $\mathcal{B}$.*

X is the central clique of $(X, \mathcal{B})$ and $(x_1, \ldots, x_t)$ a good permutation of $(X, \mathcal{B})$.

Theorem 12. *A graph G admits a good partition iff it admits a star NeS model. Moreover, given the former we can compute the latter in polynomial time.*

Proof (Sketch of proof). ($\Leftarrow$) Let $(\mathcal{T}, (T_v)_{v \in V(G)})$ be a star NeS model of a graph G and $(X, \mathcal{B})$ be the pair that we defined above Fact 8. Properties 1 and 2 follows immediately from the first two items of Fact 8. Take the permutation $(x_1, \ldots, x_t)$ defined above Definition 10. As argued there, for every $i \in [t]$, $N(x_i)$ is minimal in $\{x_i, \ldots, x_t\}$ for at least $|\mathcal{B}| - 1$ blocks of $\mathcal{B}$, and thus $(x_1, \ldots, x_t)$ is a good permutation and Property 3 is satisfied.

($\Rightarrow$) Let $(X, \mathcal{B})$ be a good partition of a graph G with $\mathcal{B} = \{B_1, \ldots, B_\beta\}$ and $(x_1, \ldots, x_t)$ be a good permutation of $(X, \mathcal{B})$. Take $\mathcal{T}$, the embedding of a star with center c that is the union of β line segments $L_1, \ldots, L_\beta$ whose intersection is $\{c\}$. We start by constructing the neighborhood subtrees of the vertices in X. For doing so, we associate each $x_i \in X$ and each line segment L_j with a rational $\ell(x_i, L_j)$ and define T_{x_i} as the union over $j \in [\beta]$ of the points on L_j at distance at most $\ell(x_i, L_j)$ from c. We set $\ell(x_i, L_j) = i$ if $N(x_i)$ is minimal in $\{x_i, \ldots, x_t\}$ for B_j, and otherwise, we set $\ell(x_i, L_j)$ to a value computed from the lengths associated with the vertices $x_{i+1}, \ldots, x_t$ on L_j such that ($\clubsuit$) for every $x, y \in X$, if $N(x) \cap B_j \subset N(y)$ then $\ell(x, L_j) < \ell(y, L_j)$.

Without going into details of how we compute these lengths, roughly what we do is the following: for each i and each j, if we have some vertices $\hat{x}$ in X "non-minimal" on B_j such that $N(x_i) \cap B_j \subseteq N(\hat{x}) \cap B_j \subset N(x_{i+1}) \cap B_j$, then we place each $\hat{x}$ spaced between i and $i+1$. Since $(x_1, \ldots, x_t)$ is a good permutation, we deduce that for each i the lengths $\ell(x_i, L_j)$ respect the last item of Fact 8 and each T_{x_i} is a neighborhood subtree. For the vertices not in X, with every $j \in [\beta]$, we associate each vertex $v \in B_j$ to an interval of L_j so that $(L_i, (T_v \cap L_i)_{v \in B_i \cup X})$ is an interval model of $G[X \cup B_i]$, which can always be done since we know that ($\clubsuit$) holds. This allows us to obtain a NeS model in polynomial time. □

It is easy to see that every graph that admits a star NeS model has a clique-tree that is a subdivided star. The converse is not true. In fact, for every graph G with a clique tree that is a subdivided star with center X, the pair $(X, \mathsf{CC}(G - X))$ satisfies Properties 1 and 2 of Definition 11 but Property 3 might not be satisfied. See for example the graph in Fig. 2 and note that the pair $(\{a, b, c\}, \mathsf{CC}(G - \{a, b, c\}))$ does not satisfy Property 3, as after removing the vertex c neither a nor b is removable from $\{a, b\}$.

We now describe Algorithm 1 that decides whether a graph G admits a good partition. Clearly G must be chordal, so we start by checking this. A chordal graph has $O(n)$ maximal cliques, and for each maximal clique X we try to construct a good partition $(X, \mathcal{A})$ of G. We start with $\mathcal{A} \leftarrow \mathsf{CC}(G - X)$ and note that $(X, \mathcal{A})$ trivially satisfies Property 1 of Definition 11. Moreover, if G admits a good partition with central clique X, then $(X, \mathcal{A})$ must satisfy Property 2 of Definition 11, and we check this in Line 3. Then, Algorithm 1 iteratively in a while loop tries to construct a good permutation $(w_1, \ldots, w_{|X|})$ of X, while possibly merging some blocks of $\mathcal{A}$ along the way so that it satisfies Property 3, or discover that there is no good partition with central clique X.

For doing so, at every iteration of the while loop, Algorithm 1 searches for a vertex w in W (the set of unprocessed vertices) such that $\mathsf{notmin}(w, W, \mathcal{A})$ – the union of the blocks of $\mathcal{A}$ where $N(w)$ is not minimal for W (see Definition 13) – induces an X-interval graph with X. If such a vertex w exists, then Algorithm 1 sets w_r to w, increments r and merges the blocks of $\mathcal{A}$ contained in $\mathsf{notmin}(w_r, W, \mathcal{A})$ (Line 8) to make w_r removable in $\mathcal{A}$ for W. Otherwise, when no such vertex w exists, Algorithm 1 stops the while loop (Line 7) and tries another candidate for X. For the graph in Fig. 2, with $X = \{a, b, c\}$ the first iteration of the while loop will succeed and set $w_1 = c$, but in the second iteration neither a nor b satisfy the condition of Line 6.

At the start of an iteration of the while loop, the algorithm has already choosen the vertices $w_1, w_2, ..., w_{r-1}$ and $W = X \setminus \{w_1, \ldots, w_{r-1}\}$. For every $i \in [r-1]$, w_i is removable from $X \setminus \{w_1, \ldots, w_{i-1}\}$ for $\mathcal{A}$. According to Definition 10 the next vertex w_r to be removed should have $N(w_r)$ non-minimal for W in at most one block of the good partition we want to construct. However, the neighborhood $N(w_r)$ may be non-minimal for W in several blocks of the current partition $\mathcal{A}$, since these blocks may be (unions of) separate components of $G \setminus X$ that should live on the same line segment of a star NeS model and thus actually be a single block which together with X induces an X-interval graph. An example of this merging is given in Fig. 3. The following definition captures, for each $w \in W$, the union of the blocks of $\mathcal{A}$ where $N(w)$ is not minimal for W.

Definition 13 (notmin). *For $W \subseteq X$, $x \in W$ and partition $\mathcal{A}$ of $V(G) \setminus X$, we denote by $\mathsf{notmin}(x, W, \mathcal{A})$ the union of the blocks $A \in \mathcal{A}$ where $N(x)$ is not minimal in A for W.*

As we already argued, when Algorithm 1 starts a while loop, $(X, \mathcal{A})$ satisfies Properties 1 and 2, and it is not hard to argue that in each iteration, for every $i \in [r-1]$, w_i is removable from $X \setminus \{w_1, \ldots, w_{i-1}\}$ for $\mathcal{A}$. Hence, when $W = \varnothing$, then $(w_1, \ldots, w_{|X|})$ is a good permutation of $(X, \mathcal{A})$ and Property 3 is satisfied.

Lemma 14. *If Algorithm 1 returns $(X, \mathcal{B})$, then $(X, \mathcal{B})$ is a good partition.*

To prove the opposite direction, namely that if G has a good partition $(X, \mathcal{B})$ associated with a good permutation $(x_1, \ldots, x_t)$, then Algorithm 1 finds a good partition, we need two lemmata. The easy case is when Algorithm 1 chooses consecutively $w_1 = x_1, \ldots, w_t = x_t$, and we can use Lemma 15 to prove that it will not return no. However, Algorithm 1 does not have this permutation as input and at some iteration with $w_1 = x_1, \ldots, w_{r-1} = x_{r-1}$, the algorithm might

Algorithm 1:

Input: A graph G.
Output: A good partition of G or "no".
1 Check if G is chordal and if so, compute its set of maximal cliques $\mathcal{X}$, otherwise **return** no;
2 **for** *every* $X \in \mathcal{X}$ **do**
3   **if** *there exists* $C \in \mathsf{CC}(G - X)$ *such that* $G[X \cup C]$ *is not an* X*-interval graph* **then continue**;
4   $\mathcal{A} \leftarrow \mathsf{CC}(G - X)$, $W \leftarrow X$ and $r \leftarrow 1$;
5   **while** $W \neq \varnothing$ **do**
6   **if** *there exists* $w \in W$ *such that* $G[X \cup \mathsf{notmin}(w_r, W, \mathcal{A})]$ *is an* X*-interval graph* **then** $w_r \leftarrow w$;
7   **else break**;
8   Replace the blocks of $\mathcal{A}$ contained in $\mathsf{notmin}(w_r, W, \mathcal{A})$ by $\mathsf{notmin}(w_r, W, \mathcal{A})$;
9     $W \leftarrow W \setminus \{w_r\}$ and $r \leftarrow r + 1$;
10   **end**
11   **if** $W = \varnothing$ **then return** $(X, \mathcal{A})$
12 **end**
13 **return** no;

stop to follow the permutation $(x_1, \ldots, x_t)$ and choose a vertex $w_r = x_i$ with $r < i$ because x_r may not be the only vertex satisfying the condition of Line 6. In Lemma 16 we show that choosing $w_r = x_i$ is then not a mistake as it implies the existence of another good partition and another good permutation that starts with $(x_1, \ldots, x_{r-1}, w_r = x_i)$. See Fig. 3 for an example of a very simple graph with several good permutations leading to quite distinct star NeS models.

We need some definitions. Given permutation $P = (x_1, \ldots, x_\ell)$ of a subset of X and $i \in [\ell]$, define $\mathcal{A}_0^P = \mathsf{CC}(G - X)$ and $\mathcal{A}_i^P$ the partition of $V(G) \setminus X$ obtained from $\mathcal{A}_{i-1}^P$ by merging the blocks contained in $\mathsf{notmin}(x_i, X \setminus \{x_1, \ldots, x_{i-1}\}, \mathcal{A}_{i-1}^P)$. Observe that when Algorithm 1 treats X and we have $w_1 = x_1, \ldots, w_\ell = x_\ell$, then the values of $\mathcal{A}$ are successively $\mathcal{A}_0^P, \ldots, \mathcal{A}_\ell^P$. The following lemma proves that if there exists a good permutation $P = (x_1, \ldots, x_t)$ and at some iteration we have $w_1 = x_1, \ldots, w_{r-1} = x_{r-1}$, then the vertex x_r satisfies the condition of Line 6 and Algorithm 1 does not return no during this iteration. Thus, as long as Algorithm 1 follows a good permutation, it will not return no.

Lemma 15. *Let G be a graph with good partition $(X, \mathcal{B})$ and $P = (x_1, \ldots, x_t)$ be a good permutation of $(X, \mathcal{B})$. For every $i \in [t]$, we have $\mathcal{A}_i^P \sqsubseteq \mathcal{B}$ and the graph $G[X \cup \mathsf{notmin}(x_i, \{x_i, \ldots, x_t\}, \mathcal{A}_{i-1}^P)]$ is an X-interval graph.*

Lemma 16. *Let $P = (x_1, \ldots, x_t)$ be a good permutation of X and $i \in [t]$. For every $w \in \{x_i, \ldots, x_t\}$ such that $G[X \cup \mathsf{notmin}(w, \{x_i, \ldots, x_t\}, \mathcal{A}_{i-1}^P)]$ is an X-interval graph, there exists a good permutation of X starting with $(x_1, \ldots, x_{i-1}, w)$.*

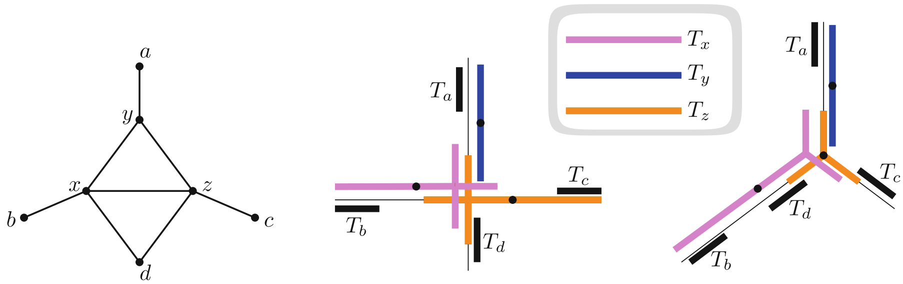

Fig. 3. A graph G and two star NeS models. The dots are the centers of the subtrees. $X = \{x, y, z\}$ is a maximal clique with four components in $G - X$. The left NeS model corresponds to permutations (y, x, z) or (y, z, x) and the right to permutation (x, y, z). If Algorithm 1 chooses $w_1 = x$, then the components $\{b\}$ and $\{d\}$ will be merged in Line 8. There is a third star NeS model, corresponding to permutation (z, y, x), similar to the one on the right. The last two permutations of X are not good permutations.

Proof. (Sketch of proof) For every $A \subseteq V(G)\backslash X$, we define $X\text{-}\mathsf{max}_\cap(A) = N(A) \cap X$ and $X\text{-}\mathsf{min}_\cap(A) = \cap_{a \in A} N(a) \cap X$. Given $A_1, A_2 \subseteq V(G)\backslash X$, we say that $A_1 \leqslant_X A_2$ if $X\text{-}\mathsf{max}_\cap(A_1) \subseteq X\text{-}\mathsf{min}_\cap(A_2)$. In Fig. 3, we have $\{b\} \leqslant_{\{x,y,z\}} \{d\}$. We denote by X_i the set $\{x_i, \dots, x_t\}$. Let $w \in X_i$ such that the graph $G[X \cup \mathsf{notmin}(w, X_i, \mathcal{A}^P_{i-1})]$ is an X-interval graph. Let $(X, \mathcal{B})$ be a good partition of G such that $(x_1, \dots, x_t)$ is a good permutation of $(X, \mathcal{B})$. If $w = x_p$ is removable from $\mathcal{B}$ for X_i, then we are done since $(x_1, \dots, x_{i-1}, w, x_i, \dots, x_{p-1}, x_{p+1}, \dots, x_t)$ is a good permutation of $(X, \mathcal{B})$. In particular, w is removable from $\mathcal{B}$ if we have $|\mathsf{notmin}(w, X_i, \mathcal{A}^P_{i-1})| \leqslant 1$.

We supose from now that $|\mathsf{notmin}(w, X_i, \mathcal{A}^P_{i-1})| \geqslant 2$. We construct a good partition $(X, \mathcal{B}_{\mathsf{new}})$ that admits a good permutation starting with $(x_1, \dots, x_{i-1}, w)$. Let $A_1, \dots, A_k$ be the blocks of $\mathcal{A}^P_{i-1}$ such that $\mathsf{notmin}(w, X_i, \mathcal{A}^P_{i-1}) = A_1 \cup A_2 \cup \dots \cup A_k$. We prove a separate claim implying that as $G[X \cup \mathsf{notmin}(w, X_i, \mathcal{A}^P_{i-1})]$ is an X-interval graph, the blocks $A_1, \dots, A_k$ are pairwise comparable for $\leqslant_X$. Suppose w.l.o.g. that $A_1 \leqslant_X \dots \leqslant_X A_k$.

In Fig. 3 with $\mathsf{notmin}(x, \{x, y, z\}, \{\{a\}, \{b\}, \{c\}, \{d\}\}) = \{b\} \cup \{d\}$ we have $A_1 = \{b\}$ and $A_2 = \{d\}$. By Lemma 15, we have $\mathcal{A}^P_{i-1} \sqsubseteq \mathcal{B}$. Thus, there exists a block B_1 of $\mathcal{B}$ containing A_1 and B_1 is a union of blocks of $\mathcal{A}^P_{i-1}$. See Fig. 4. Let B_1^{max} be the union of all the blocks A of $\mathcal{A}^P_{i-1}$ included in B_1 such that A is not contained in $\mathsf{notmin}(w, X_i, \mathcal{A}^P_{i-1})$ and $A_1 \leqslant_X A$. Note that for every block $A \in B_1^{\mathsf{max}}$, we have $A_k \leqslant_X A$ because otherwise we have $A_1 \leqslant_X A \leqslant_X A_k$ and that implies $A \subseteq \mathsf{notmin}(w, X_i, \mathcal{A}^P_{i-1})$.

We prove a separate claim allowing us to assume that $B_1^{\mathsf{max}} = \varnothing$. We then construct the partition $\mathcal{B}_{\mathsf{new}}$ as follows. Recall that $A_1 \subseteq B_1$. We create a new block $\widehat{B}_1 = B_1 \cup A_2 \cup \dots \cup A_k$, and for every block $B \in \mathcal{B}$ such that $B \neq B_1$, we create a new block $\widehat{B} = B \backslash (A_2 \cup \dots \cup A_k)$. We define $\mathcal{B}_{\mathsf{new}} = \{\widehat{B}_1\} \cup \{\widehat{B} : B \in \mathcal{B}\backslash\{B_1\}$ and $\widehat{B} \neq \varnothing\}$. The construction of $\mathcal{B}_{\mathsf{new}}$ is illustrated in Fig. 4. Finally, we

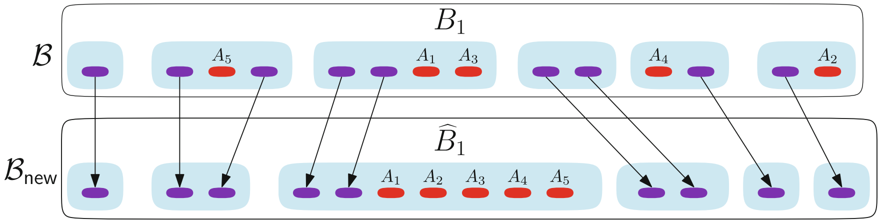

Fig. 4. Construction of $\mathcal{B}_{\mathsf{new}}$ from $\mathcal{B}$ with $k = 5$. The blocks of $\mathcal{B}$ and $\mathcal{B}_{\mathsf{new}}$ are in blue, the blocks $A_1, \ldots, A_5$ of $\mathcal{A}^P_{i-1}$ contained in $\mathsf{notmin}(w, X_i, \mathcal{A}^P_{i-1})$ are in red, the other blocks of $\mathcal{A}^P_{i-1}$ are in purple. In each block of $\mathcal{B}$ or $\mathcal{B}_{\mathsf{new}}$, the blocks of $\mathcal{A}^P_{i-1}$ are ordered w.r.t. $\leqslant_X$ from left to right. (Color figure online)

prove 4 separate claims that allow us to show that $(X, \mathcal{B}_{\mathsf{new}})$ is a good partition for a good permutation starting with $(x_1, \ldots, x_{i-1}, w)$. □

Lemma 17. *If G has a good partition then Algorithm 1 returns one.*

Proof. Suppose G admits a good partition with central clique X. We prove the following invariant holds at the end of the i-th iteration of the while loop for X.

Invariant. G admits a good permutation starting with $w_1, ..., w_i$.

By assumption, X admits a good permutation and thus the invariant holds before the algorithm starts the first iteration of the while loop. By induction, assume the invariant holds when Algorithm 1 starts the i-th iteration of the while loop. Let $L = (w_1, \ldots, w_{i-1})$ be the consecutive vertices chosen at Line 6 before the start of iteration i (observe that L is empty when $i = 1$). The invariant implies that there exists a good permutation $P = (w_1, \ldots, w_{i-1}, x_i, x_{i+1}, ..., x_t)$ of G starting with L. By Lemma 15, the graph $G[X \cup \mathsf{notmin}(x_i, \{x_i, \ldots, x_t\}, \mathcal{A}^P_{i-1})]$ is an X-interval graph. Observe that $\mathcal{A}^P_{i-1}$ and $\{x_i, \ldots, x_t\}$ are the values of the variables $\mathcal{A}$ and W when Algorithm 1 starts the i-th iteration. Thus, at the start of the i-th iteration, there exists a vertex w_i such that $G[X \cup \mathsf{notmin}(w_i, W, \mathcal{A})]$ is an X-interval graph. Consequently, the algorithm does not return no at the i-th iteration and chooses a vertex $w_i \in \{x_i, \ldots, x_t\}$ such that the graph $G[X \cup \mathsf{notmin}(w_i, \{x_i, \ldots, x_t\}, \mathcal{A}^P_i)]$ is an X-interval graph. By Lemma 16, G admits a good permutation starting with $(w_1, \ldots, w_{i-1}, w_i)$. Thus, the invariant holds at the end of the i-th iteration. If at the end of the i-th iteration W is empty, then the while loop stops and the algorithm returns a pair $(X, \mathcal{A})$. Otherwise, the algorithm starts an $i + 1$-st iteration and the invariant holds at the start of this new iteration. By induction, the invariant holds at every step and Algorithm 1 returns a pair $(X, \mathcal{A})$ which is a good partition by Lemma 14. □

Theorem 18. *Algorithm 1 decides in polynomial time if G admits a star NeS model.*

Proof. By Theorem 12 G admits a star NeS model iff G admits a good partition, and by Lemma 14 and 17 Algorithm 1 finds a good partition iff the input graph has a good partition. Let us argue for the runtime. Checking that G is chordal and finding the $O(n)$ maximal cliques can be done in polynomial time [16,23]. Given $X, Y \subseteq V(G)$ we check whether $G[X \cup Y]$ is an X-interval graph, as follows. Take G' the graph obtained from $G[X \cup Y]$ by adding u and v such that $N(u) = \{v\}$ and $N(v) = \{u\} \cup X$. It is easy to see that $G[X \cup Y]$ is an X-interval graph iff G' is an interval graph, which can be checked in polynomial time. $\square$

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